

Deterministic Lattice-Brane Substrate in a 4D Embedding

Emergent Lorentz Kinematics, Gauge Holonomy, and Solitonic Matter

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Abstract

The Standard Model and general relativity provide exceptionally accurate effective descriptions of observed physics, yet their combined “fundamental” inventory is structurally elaborate: multiple quantum fields, symmetry sectors, and many parameters fixed by measurement. This motivates a complementary question of *minimal-ingredient reconstruction*: can a smaller set of ground ingredients plausibly give rise to the familiar phenomenology as emergent, coarse-grained regularities?

We explore a deterministic *substrate-first* framework in which observable physics arises from waves on an optically real, tensioned three-dimensional elastic medium (a “brane”) embedded in $\mathbb{R}^4$. The continuum model is formulated for an embedding field $\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4$ with external time as evolution parameter, using an isotropic hyperelastic action built from the induced metric and an isotropic reference metric; a single mismatch parameter α encodes homogeneous pre-stress. The continuum description is treated as the *long-wavelength effective* limit of a cubic lattice of nodes/cells embedded in $\mathbb{R}^4$. Linearization yields multiple polarized branches with characteristic wave cones; when a single branch dominates and dispersion is negligible, Lorentz-type kinematics emerges as an effective symmetry for observers built from the same substrate. A further geometric consequence of embedding is a transverse–lateral backreaction: localized excitation of the fourth component changes intrinsic distances and, under pre-tension, drives an inward lateral pull (contraction) that acts as a long-range “gravity-like” channel in a weak, slowly varying limit. We additionally postulate an ordered, narrowband carrier sector that provides macroscopic phase coherence. Adiabatic transport of its polarization subspace induces a Berry (rank-1) or Wilczek–Zee (non-Abelian) connection; we interpret these connections as the natural gauge degrees of freedom of a coarse-grained envelope description, leading to gauge-covariant derivatives and curvature terms in an effective slow-sector action. On a cubic lattice, a natural internal sector is a three-component axis-aligned narrowband triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$. In the weak-mixing limit $g_{\text{mix}} \rightarrow 0$, each axis supports an independent $U(1)$ geometric phase. Here g_{mix} is an effective measure of inter-axis mixing controlled by the prestretch α and by the microscopic coupling stencil (e.g. diagonal couplings). For $g_{\text{mix}} > 0$, the eigenmodes become mixtures of axes and the correct description is a non-Abelian $U(3)$ Wilczek–Zee connection whose traceless part suggests an $SU(3)$ -like “color” structure. The lattice spacing also introduces a Brillouin-zone setting in which Berry curvature can be computed on a discretized momentum lattice, in direct analogy with lattice-QCD Berry-curvature constructions. Particle-like states are modeled as localized solitons: because the dynamics derives from an action, linearization about symmetric backgrounds yields self-adjoint eigenproblems, with angular structure organized by spherical-harmonic families and physical amplitudes fixed by nonlinear self-guidance and energy normalization. A discrete lattice realization is recorded both as (i) the proposed microphysical substrate picture and (ii) an implementation basis for testing dispersion, holonomy diagnostics, and localized-mode stability.

The aim is not to challenge the empirical success of established theories, but to present a concrete, testable substrate model in which “matter” and “forces” are different regimes of one underlying dynamics.

1 Introduction

The Standard Model and general relativity are extraordinarily successful at organizing experimental facts. At the same time, their combined fundamental “catalog”—multiple fields, symmetry sectors, symmetry breaking structure, and many parameters fixed by measurement—invites a complementary line of inquiry: whether a smaller set of ground ingredients could underlie this effective complexity. This paper explores such a *minimal-ingredient reconstruction* approach.

Substrate-first strategy. We adopt a deterministic *substrate-first* strategy: posit an ontically real wave medium and treat familiar field theories as coarse-grained descriptions of constrained wave dynamics. The goal is *not* to dispute the established phenomenology of quantum field theory, but to ask what could make gauge structure, relativistic kinematics, and particle-like quantization appear naturally as emergent regularities.

Ground ingredients (minimal commitments). The theory fragment developed here aims to keep its primitive commitments small: (i) a single substrate with local action dynamics (posited microscopically as a cubic lattice embedded in $\mathbb{R}^4$, with a continuum long-wavelength effective description), (ii) geometric nonlinearity arising from the embedding (rather than adding separate interaction fields by hand), and (iii) a controlled separation between a linear wave sector and a nonlinear self-guidance sector. Symmetry properties such as large-scale isotropy are treated as *constraints on admissible constitutive choices* (e.g., writing the energy in terms of invariants of the induced metric), not as additional “ingredients”. Everything else is treated as emergent bookkeeping: in particular, gauge potentials are interpreted as connection data of an ordered wave sector, not as independent fundamental substances.

Substrate hypothesis. We assume a three-dimensional elastic medium (“brane”) embedded in $\mathbb{R}^4$, described by an embedding map

$$\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4, \quad (x, t) \mapsto \mathbf{X}(x, t), \quad \Omega \subset \mathbb{R}^3, \quad (1)$$

with an external time parameter t . The dynamics is specified by a local hyperelastic action on Ω , constructed from invariants of the induced metric of the embedding. This is the minimal coordinate-free way to write an elastic energy without committing the effective continuum description to a preferred direction; microscopically, a cubic lattice substrate is allowed, and large-scale isotropy is treated as an empirical constraint on admissible lattice couplings and parameter regimes [5, 7, 10].

Gravity from transverse excitation (contraction). A central motivating idea—already present at the level of the induced metric—is that excitation in the embedding dimension generically feeds back into *lateral* stresses. Intuitively, if a region of the brane develops transverse structure $X^4 = u(x, t)$, then the induced metric gains a quadratic contribution $\partial_i u \partial_j u$; because the in-brane displacement is dynamical, a pre-tensioned medium relaxes by generating an inward-biased lateral response rather than only a local stress increase. Coarse-grained over scales large compared to the carrier wavelength, such slowly varying contraction fields can bias the propagation of other wave packets and the equilibrium of localized modes. We therefore treat Newtonian gravity as an emergent, long-range *contraction channel* mediated by transverse excitation, not as a separately postulated fundamental field.

Narrowband order and geometric phase. A further postulate is the existence of an ordered, narrowband carrier sector that provides a coherent phase reference over macroscopic scales (assumed to live primarily in a longitudinal polarization branch). On a lattice substrate,

band structure provides a natural setting for narrowband sectors; here we treat the existence of an ordered carrier sector as a working postulate and focus on its diagnostic consequences. Geometric phase and holonomy are universal in wave systems and in adiabatic transport of polarized states [2–4, 11, 15]. Here they are promoted to a structural bridge: adiabatic transport of a polarization subspace induces a Berry (rank-1) or Wilczek–Zee (non-Abelian) connection, which is interpreted as the natural origin of gauge degrees of freedom in an effective description.

Particles as solitons; probability not fundamental. We do not take the probabilistic interpretation of quantum theory as fundamental. Instead, apparent point-like interactions and discrete outcomes are attributed to deterministic wave dynamics combined with Fourier-localization constraints and threshold-like nonlinear exchange events. Particle-like states are localized solitons whose spatial structure is organized by spherical symmetry and spherical-harmonic mode families. The spin- $\frac{1}{2}$ 4π aspect is attributed to geometric phase/holonomy of an internal polarization subspace (a “spinorial” parallel transport effect) rather than to literal multi-winding of a photon path.

Relation to existing ideas. The emergence of an effective light cone and Lorentz-like kinematics from a preferred-time medium has close analogues in analogue-gravity systems [1, 9, 14]. The present use of the term “brane” is literal (an elastic medium), and should not be confused with field-theoretic brane-world scenarios in high-energy theory [13].

Paper organization. Section 2 states scope, contributions, and non-claims. Section 3 formulates the continuum substrate model (written in a coordinate-free form with an isotropic constitutive baseline). Section 4 discusses the linear-wave regime and an emergent effective light cone. Sections 5 and 6 develop the geometric connection picture (Berry/Wilczek–Zee) and its gauge-covariant envelope formulation. Section 7 treats localized modes and shows how to pose them as self-adjoint eigenproblems around symmetric backgrounds. The microphysical lattice postulate and its parameterization are stated in Section 3.2; geometric-phase diagnostics are developed in Section 5.

2 Scope, contributions, and non-claims

This paper develops a *deterministic substrate theory fragment* with an explicit goal: to formulate a coordinate-free continuum model (using an isotropic constitutive baseline) with clear points of contact to simulations and diagnostics, and to make the remaining open problems visible. A *cubic lattice* is treated as a candidate *microphysical* substrate, while the continuum formulation is retained as the *long-wavelength effective description* and as a compact way to express isotropic constitutive baselines.

2.1 Positioning relative to established physics

The Standard Model and general relativity are treated here as *empirically validated effective frameworks*. Nothing in this paper is intended to reduce confidence in their experimental domain of applicability. The present work instead targets a different explanatory layer: it asks whether familiar structures such as effective Lorentz symmetry, gauge connections, and quantized particle spectra could arise as emergent regularities from a smaller set of ground ingredients.

2.2 Core building blocks

The theory fragment developed here is organized around six coupled elements:

1. **Lattice substrate and continuum limit.** A cubic lattice of nodes/cells embedded in $\mathbb{R}^4$ is taken as a candidate microphysical substrate. For wavelengths large compared to the lattice spacing, the dynamics admits a continuum approximation by an isotropic hyperelastic action on an intrinsic 3-domain $\Omega \subset \mathbb{R}^3$ (Section 3).
2. **Continuum 3D brane in 4D embedding (effective description).** A local hyperelastic action on an intrinsic 3-manifold $\Omega \subset \mathbb{R}^3$ (taken in an isotropic baseline form, i.e. without built-in preferred directions) determines the dynamics of an embedding $\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4$ (Section 3).
3. **Gravity as contraction from transverse excitation.** Because the brane is embedded, transverse gradients in X^4 change intrinsic distances through the induced metric. Under homogeneous pre-tension ($\alpha < 1$), localized transverse excitation therefore produces an inward lateral pull (contraction) and a slowly varying background strain field. In a weak, slowly varying regime we interpret this transverse–lateral backreaction as the qualitative origin of a long-range gravity-like channel (Section 3.8).
4. **Ordered narrowband carrier sector.** We assume the substrate admits an ordered, narrowband carrier sector (taken to be primarily longitudinal) that provides phase rigidity and enables a controlled multiscale envelope description (Sections 5 and 6).
5. **Berry/Wilczek–Zee connections as gauge structure.** Adiabatic transport of a polarization subspace induces a Berry connection (rank-1) or a Wilczek–Zee connection (non-Abelian). These connections are interpreted as the natural origin of gauge degrees of freedom in a coarse-grained description, and provide the bridge to a $U(1)$ -like gauge picture and its non-Abelian extension (Sections 5 and 6).
6. **Axis-aligned triplet and coupling-induced mixing (toward $SU(3)$).** On a cubic substrate, an axis-aligned narrowband triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ is natural. For $g_{\text{mix}} = 0$ the three phases behave as $U(1) \times U(1) \times U(1)$; for $g_{\text{mix}} > 0$ eigenmodes mix and the correct adiabatic transport is a $U(3)$ Wilczek–Zee connection (with a traceless $SU(3)$ part).
7. **Solitonic particles and a Dirac-envelope split.** Particle-like states are localized solitons whose spatial structure is organized by spherical symmetry and spherical-harmonic mode families (Section 7). Conceptually, we separate (i) particle-internal structure (topology/geometry and its internal labels, including matter/antimatter distinction and spin state) from (ii) the underlying local phase structure of the carrier sector. The standard Dirac equation mixes both; here it is treated as an effective envelope description that couples the particle sector to the local phase/gauge structure. The spin- $\frac{1}{2}$ 4π aspect is attributed to geometric phase/holonomy of an internal polarization subspace.

2.3 Non-claims (what is *not* delivered here)

To keep the scope honest, several items are explicitly *not* claimed in this version:

- a derivation of the full Standard Model gauge group, generations, or precise coupling constants,
- a completed quantitative reduction of the contraction channel to a Newtonian limit with a derived value of G ,
- a complete derivation of Maxwell/Yang–Mills dynamics for the emergent connection in all regimes,
- a closed multi-particle theory including scattering cross sections, bound-state spectra, and entanglement phenomenology.

2.4 Falsifiable signatures and diagnostic targets

The framework makes concrete predictions that can be tested (and potentially ruled out) in numerical experiments on the lattice substrate and in any future phenomenological matching. Key diagnostic targets include:

- **Lorentz-violation proxies:** direction-dependent wave speeds, birefringence, and dispersion onset as functions of wavelength relative to lattice spacing.
- **Branch leakage / adiabaticity failure:** measurable conversion between polarization branches when the gap condition Equation (28) is violated.
- **Gauge-holonomy invariants:** gauge-invariant loop observables (rank-1 phase; rank- n holonomy spectrum) extracted by the protocol in Section 5.4.
- **Non-Abelianity:** noncommuting holonomies for distinct loops in regimes where a multi-dimensional transported subspace is retained.
- **Gravity-channel response:** a slow contraction/dilation field sourced by transverse excitation and its influence on test-wave propagation in weak, slowly varying limits.
- **Localized-mode stability:** long-time persistence and perturbation response of soliton-like states (or clear failure modes if they do not exist in the chosen constitutive/stencil class).

The contribution is therefore a well-specified substrate model plus concrete diagnostics and simulation checks that can falsify or refine the proposed emergence mechanisms.

3 Substrate model: lattice microphysics and continuum (IR) effective description

3.1 Continuum long-wavelength limit: ontology and kinematics

Let $\Omega \subset \mathbb{R}^3$ be the intrinsic (material) brane domain. We use either a periodic cell $\Omega = \mathbb{T}^3$ or a bounded cube with clamped boundary points. Time $t \in \mathbb{R}$ is an external evolution parameter, not an additional coordinate of the brane.

The fundamental field is the embedding map

$$\mathbf{X}(x, t) \in \mathbb{R}^4, \quad x = (x^1, x^2, x^3) \in \Omega, \quad t \in \mathbb{R}, \quad (2)$$

whose components $X^A(x, t)$ ($A = 1, \dots, 4$) give the brane configuration in the ambient space.

3.2 Discrete lattice substrate (microphysical postulate)

The continuum model is treated as the long-wavelength effective theory of a *cubic lattice* substrate. Let lattice sites be labeled by $n \in \mathbb{Z}^3$ with spacing a and (reference) material positions $x_n = a n$. Each site carries an embedded configuration $\mathbf{X}_n(t) \in \mathbb{R}^4$. Interactions are encoded by springs/couplings across neighbor shells with tunable strengths: nearest neighbors (axis-aligned) with stiffness k_1 , face diagonals with k_2 , and body diagonals with k_3 . The rest lengths follow the geometry $(a, \sqrt{2}a, \sqrt{3}a)$ with a uniform prestretch parameter α . The ratios $(k_2/k_1, k_3/k_1)$ can be tuned to suppress leading anisotropic terms in the small- $|k|$ expansion.

Dispersion, anisotropy, and birefringence. At wavelengths comparable to a , the lattice generically exhibits direction-dependent dispersion and branch splitting (birefringence), because the cubic symmetry is not fully rotational. In contrast, for $\lambda \gg a$ (small $|k|$ inside the Brillouin zone), the dispersion can be expanded and the leading-order dynamics is well-approximated by an isotropic continuum elasticity model with effective Lamé-type parameters. This motivates using an isotropic hyperelastic action as a baseline in the main text while keeping the lattice as the microphysical hypothesis.

Brillouin zone and geometric-phase diagnostics. The lattice spacing a defines a Brillouin zone $k_i \in [-\pi/a, \pi/a]$. Because Berry curvature is naturally defined over parameter spaces such as $\mathbf{k}$ -space eigenbundles, the lattice formulation provides a direct route to Berry-curvature diagnostics on a discretized momentum lattice (see Section 5).

3.3 Induced metric and isotropic reference metric

The embedding induces a (time-dependent) Riemannian metric on the brane:

$$g_{ij}(x, t) := \partial_i \mathbf{X}(x, t) \cdot \partial_j \mathbf{X}(x, t), \quad i, j = 1, 2, 3. \quad (3)$$

This definition is invariant under rigid motions $\mathbf{X} \mapsto Q\mathbf{X} + a$ with $Q \in O(4)$ and $a \in \mathbb{R}^4$. Rotational invariance within the brane is obtained by constructing the stored energy density from invariants of g relative to an isotropic reference metric.

We choose a homogeneous, isotropic reference metric

$$g_{ij}^0 = \ell_0^2 \delta_{ij}, \quad (4)$$

where $\ell_0 > 0$ is the continuum analogue of a discrete spring rest length. Define the mixed relative metric

$$\hat{g} := (g^0)^{-1} g. \quad (5)$$

Any isotropic hyperelastic energy density can be written as $W = W(\hat{g})$, equivalently as a function of the principal stretches or invariants of $\hat{g}$.

Continuum analogue of pre-tension. The reference metric g^0 encodes the stress-free geometric scale. If boundary conditions enforce a held ground-state scale $h_\star$ (uniform), then

$$\alpha := \frac{\ell_0}{h_\star} \in (0, 1] \quad (6)$$

measures the mismatch between stress-free scale and held scale. For $\alpha < 1$ the brane is uniformly pre-stressed (continuum “pre-tension”). This mismatch controls how strongly transverse gradients backreact on in-brane stresses through the geometric nonlinearity of $g_{ij}(\mathbf{X})$. In the lattice substrate picture (Section 3.2), the same α multiplies the geometric neighbor distances to set spring rest lengths; Equation (6) is the continuum encoding of that uniform prestretch.

3.4 Constitutive law: St. Venant–Kirchhoff (Green–Lagrange strain)

As a concrete baseline we use the St. Venant–Kirchhoff (StVK) energy expressed in terms of the Green–Lagrange strain tensor

$$E := \frac{1}{2}(\hat{g} - I) \quad (\text{with } I \text{ the } 3 \times 3 \text{ identity}). \quad (7)$$

The StVK stored energy density is

$$W_{\text{StVK}}(E) = \mu \operatorname{tr}(E^2) + \frac{\lambda}{2} (\operatorname{tr} E)^2, \quad (8)$$

with Lamé parameters $\mu > 0$ and λ (units: Pa). In terms of Poisson ratio $\nu \in (0, 1/2)$,

$$\nu = \frac{\lambda}{2(\lambda + \mu)}, \quad \lambda = \frac{2\mu\nu}{1 - 2\nu}. \quad (9)$$

This choice is not claimed unique; it is used because it is isotropic, analytic, and captures the geometric nonlinearity through $g_{ij}(\mathbf{X})$. It also reduces to standard linear elasticity in the small-strain limit, making the longitudinal/transverse wave speeds and the linearized branch structure transparent. We emphasize that StVK is employed here as a *baseline constitutive law* for analytic tractability, not as a claim of material realism at arbitrarily large strains; more robust hyperelastic energies (e.g. neo-Hookean, Mooney–Rivlin, Ogden-type) can be substituted within the same induced-metric framework without changing the geometric mechanism of transverse–lateral backreaction.

3.5 Lagrangian, action, and equations of motion

Let $\rho_m > 0$ be the brane mass density (kg/m³). The Lagrangian density is

$$\mathcal{L}(\mathbf{X}, \partial_t \mathbf{X}, \nabla \mathbf{X}) = \frac{\rho_m}{2} |\partial_t \mathbf{X}|^2 - W(g(\mathbf{X}); g^0). \quad (10)$$

The action is

$$S[\mathbf{X}] = \int_{\mathbb{R}} dt \int_{\Omega} d^3x \mathcal{L}. \quad (11)$$

Define the first Piola stress (as a momentum conjugate to $\partial_i X^A$)

$$P_A^i := \frac{\partial W}{\partial(\partial_i X^A)}. \quad (12)$$

For energies depending on g_{ij} one can write $P_A^i = 2S^{ij}\partial_j X^A$ with a symmetric second Piola-type tensor $S^{ij} = \partial W / \partial g_{ij}$. The Euler–Lagrange equations take divergence form

$$\rho_m \partial_{tt} X^A = \partial_i P_A^i, \quad A = 1, \dots, 4, \quad (13)$$

showing explicitly that isotropy (absence of built-in preferred directions) follows from writing W in terms of invariants of the induced metric relative to the reference metric g^0 .

3.6 Spherical material coordinates (coordinate change only)

Important: this is only a coordinate change. All dynamical statements remain those of the coordinate-free formulation $\mathbf{X} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^4$ with induced metric $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$. We introduce spherical coordinates on the *intrinsic/material* domain as a convenient chart for localized excitations; this does *not* constrain the embedding $\mathbf{X}$.

Let (r, θ, φ) be the standard spherical chart on (a subset of) $\mathbb{R}^3$, with Jacobian $J(r, \theta, \varphi) = r^2 \sin \theta$. Writing the intrinsic coordinate as $q^a \in \{r, \theta, \varphi\}$ and denoting derivatives in this chart by ∂_a , the equations of motion become

$$\rho_m \partial_{tt} X^A = \frac{1}{J(q)} \partial_a (J(q) P_A^a), \quad A = 1, \dots, 4, \quad (14)$$

which is exactly the Cartesian divergence form (13) written with the standard Jacobian factor.

3.7 Linearization and polarized branches

Linearize around a homogeneous background embedding $\mathbf{X}_\star(x) = (x, 0)$ and small displacements $\mathbf{u}(x, t) = \mathbf{X}(x, t) - \mathbf{X}_\star(x)$. Under standard regularity assumptions on W , one obtains a linear wave system with (in general) multiple polarization branches, including transverse and longitudinal modes. For later use, we emphasize two generic features: (i) different branches may have different characteristic speeds (and may become dispersive at shorter wavelengths), and (ii) for $\alpha < 1$ the pre-stress couples transverse excursions (in the embedding) back into in-brane stresses through the induced metric (3). These branches provide the substrate carriers for the geometric-phase and soliton constructions developed below.

3.8 Transverse–lateral coupling and an emergent “gravity” channel

A distinctive feature of an embedded elastic medium is that transverse gradients change intrinsic distances. To make this explicit without removing in-brane dynamics, we use a near-identity decomposition around the homogeneous background embedding $\mathbf{X}_\star(x) = (h_\star x, 0)$:

$$\mathbf{X}(x, t) = (h_\star x^1 + \xi^1(x, t), h_\star x^2 + \xi^2(x, t), h_\star x^3 + \xi^3(x, t), u(x, t)), \quad \vec{\xi}(x, t) \in \mathbb{R}^3. \quad (15)$$

The induced metric is then

$$\begin{aligned} g_{ij} &= \partial_i \mathbf{X} \cdot \partial_j \mathbf{X} \\ &= (h_\star \delta_i^a + \partial_i \xi^a)(h_\star \delta_j^a + \partial_j \xi^a) + \partial_i u \partial_j u \\ &= h_\star^2 \delta_{ij} + h_\star (\partial_i \xi_j + \partial_j \xi_i) + \partial_i \vec{\xi} \cdot \partial_j \vec{\xi} + \partial_i u \partial_j u. \end{aligned} \quad (16)$$

Setting $\vec{\xi} \equiv 0$ recovers the familiar graph picture used only for intuition; the “gravity” channel, however, relies on the allowed in-brane relaxation $\vec{\xi} \neq 0$ sourced by the quadratic term $\partial_i u \partial_j u$.

In the small-slope regime, the in-brane strain has the schematic form

$$\varepsilon_{ij} \approx \frac{1}{2}(\partial_i \xi_j + \partial_j \xi_i) + \frac{1}{2} \partial_i u \partial_j u, \quad (17)$$

so the quasi-static balance $\partial_i \sigma_{ij} \approx 0$ with $\sigma_{ij} = \lambda \text{tr}(\varepsilon) \delta_{ij} + 2\mu \varepsilon_{ij}$ makes it explicit that the term $\partial_i u \partial_j u$ acts as a forcing/eigenstrain-like source for the slow in-brane field $\vec{\xi}$.

Pythagorean lengthening and lateral contraction (informal picture). For a small material displacement dx , the ambient line element of the graph embedding satisfies

$$ds^2 = h_\star^2 |dx|^2 + (\nabla u \cdot dx)^2. \quad (18)$$

Thus any region with nonzero slope ∇u necessarily *lengthens* intrinsic distances compared to the flat configuration. In a discrete spring-lattice realization this is the familiar Pythagorean effect $|\Delta \mathbf{X}| = \sqrt{|\Delta x|^2 + (\Delta u)^2}$: transverse variation increases spring length even when lateral coordinates are unchanged. In the actual dynamics the in-brane displacement $\vec{\xi}$ is not constrained; under pre-tension ($\alpha < 1$) the additional metric contribution $\partial_i u \partial_j u$ acts as an effective source in the in-brane equilibrium equations, producing a compensating contraction/dilation pattern rather than a purely local increase of tensile stress. If the medium is pre-tensioned ($\alpha < 1$), the additional length is paid for by increased tensile stress, which produces net lateral forces that pull material points *toward* the region of transverse excitation. For a localized bump or rotating transverse pattern, the surrounding brane therefore develops an inward bias (contraction) rather than an outward pressure.

In this qualitative sense, the “gravitational mass” of an excitation is tied to the elastic energy stored in transverse gradients (and, more generally, to embedded deformation energy), while the “gravitational field” is the slowly varying in-brane strain/dilation pattern sourced by that

energy. The sign is fixed by geometry: increasing transverse arc length increases elastic energy and therefore increases tension, which pulls laterally inward.

With homogeneous pre-stress ($\alpha < 1$), this geometrically induced strain backreacts on the in-brane stress field through the constitutive law, effectively coupling transverse excitation energy into slow modulations of the in-brane configuration.

This provides a concrete mechanism for a long-range channel often described informally as “gravity” in the present program: localized transverse gradients can act as sources of slowly varying background strain (contraction/dilation), and linear wave packets propagating on top of that background experience branch-dependent effective coefficients (cf. Section 4). In a weak-field, slowly varying regime one may therefore seek a Newtonian-limit reduction in which a scalar potential built from coarse-grained strain (or elastic energy density) obeys a Poisson-type equation. Deriving such a reduction quantitatively (and relating the resulting coupling constant to G) is an open problem of the present theory fragment.

4 Emergent relativity and effective Lorentz symmetry

The continuum model is written with an external time parameter t , so Lorentz invariance is not postulated microscopically. Nevertheless, in the *linear wave regime* an effective Lorentz symmetry can arise as a kinematic symmetry of the dominant dispersion branch that controls information transport in a given regime.

4.1 Linear-wave cone and an effective light speed

Linearizing Equation (13) about a homogeneous background yields a multi-branch wave system. For isotropic elastic media, one expects (at long wavelengths) transverse and longitudinal branches. In the StVK baseline around a homogeneous state, the characteristic speeds take the familiar form

$$c_T^2 \sim \frac{\mu}{\rho_m}, \quad c_L^2 \sim \frac{\lambda + 2\mu}{\rho_m}, \quad (19)$$

up to the details of the chosen background and the pre-stress encoded by α . More generally, for each polarization branch b one has the long-wavelength dispersion

$$\omega_b^2(\mathbf{k}) \approx c_b^2 |\mathbf{k}|^2 \quad \text{for } |\mathbf{k}| \rightarrow 0, \quad (20)$$

which defines an effective wave cone for that branch.

In regimes where a single branch dominates transport and dispersion is negligible, the wave equation is approximately invariant under Lorentz transformations built from the corresponding speed c_b . In particular, for the dominant branch (denote its speed by c) one obtains the familiar kinematic relations for boosted frames,

$$t' = \gamma \left(t - \frac{vx}{c^2} \right), \quad x' = \gamma(x - vt), \quad \gamma = (1 - v^2/c^2)^{-1/2}, \quad (21)$$

as an effective symmetry of the linear propagation sector, not as a microscopic postulate.

4.2 Analogue-metric viewpoint

An equivalent viewpoint, common in analogue-gravity systems, is that linear excitations experience an effective metric structure with interval

$$ds^2 = -c^2 dt^2 + g_{ij}^{(\text{eff})}(x, t) dx^i dx^j, \quad (22)$$

where $g_{ij}^{(\text{eff})}$ depends on the background configuration about which one linearizes. In the present model there are two natural geometric objects: the induced brane metric $g_{ij}(\mathbf{X})$ and the

branch-dependent effective coefficients of the linearized operator. In particular, slowly varying transverse profiles $u = X^4$ modify intrinsic distances already at the level of the induced metric (Equation (16) and section 3.8), and can therefore be read as a background that focuses or defocuses wave transport in close analogy to a weak gravitational field. Both encode how background deformations can bend or modulate wave propagation, in close analogy to analogue-gravity constructions [1, 14].

4.3 Regime of validity and expected deviations

Effective Lorentz symmetry is expected to break when: (i) multiple branches contribute comparably (mode mixing), (ii) dispersion becomes significant at shorter wavelengths, or (iii) geometric/material nonlinearity becomes important at larger amplitudes. In the present theory these deviations are not treated as flaws; they are structural consequences of a preferred-time substrate and therefore delimit the domain where a relativistic effective description is expected to apply.

5 Geometric phase, Berry/Wilczek–Zee connections, and gauge structure

5.1 Ordered narrowband waves and polarization transport

We assume the substrate admits an ordered, narrowband carrier sector that supports multi-component polarized waves with a long phase-coherence length. Locally, a narrowband excitation can be written as

$$\mathbf{u}(x, t) \approx \text{Re}\left\{a(x, t) \boldsymbol{\epsilon}(x, t) e^{i\theta(x, t)}\right\}, \quad \theta(x, t) = \phi(x, t) - \omega_0 t, \quad (23)$$

where $a(x, t) \geq 0$ varies slowly, $\boldsymbol{\epsilon}(x, t) \in \mathbb{C}^N$ is a normalized polarization state ($\boldsymbol{\epsilon}^\dagger \boldsymbol{\epsilon} = 1$), and ω_0 is the carrier angular frequency. The internal dimension N counts the relevant components (e.g. embedding-direction components and/or a coarse-grained mode basis).

A central point is that polarization transport is not unique: the same physical state can be represented in many local phase conventions. Geometric phase formalizes this redundancy as a connection on a bundle of polarization states.

5.2 Rank-1 Berry connection (U(1))

For a single isolated polarization mode, represent the local state by a normalized vector $|u(x)\rangle \in \mathbb{C}^N$ that depends smoothly on spacetime position x^μ (here $x^\mu = (ct, x^1, x^2, x^3)$ built from the intrinsic coordinates and a chosen characteristic speed c of the carrier branch). Define the Berry connection

$$a_\mu(x) := i \langle u(x) | \partial_\mu u(x) \rangle. \quad (24)$$

Under a local phase choice $|u(x)\rangle \mapsto e^{-i\chi(x)} |u(x)\rangle$ one finds

$$a_\mu \mapsto a_\mu + \partial_\mu \chi, \quad (25)$$

so only gauge-invariant quantities carry physical meaning. The Berry curvature is the field strength

$$f_{\mu\nu} := \partial_\mu a_\nu - \partial_\nu a_\mu, \quad (26)$$

and the holonomy (geometric phase) around a closed loop Γ is

$$\gamma_\Gamma = \oint_\Gamma a_\mu dx^\mu = \int_{\Sigma: \partial\Sigma=\Gamma} f_{\mu\nu} d\Sigma^{\mu\nu}. \quad (27)$$

5.3 What the geometric phase is “of”: eigenmode bundles and adiabaticity

The connection above is not attached to an arbitrary decomposition of a real field into amplitude and phase. In the present setting one should choose $|u(x)\rangle$ as a (locally defined) eigenvector of the linearized carrier operator about the slowly varying background configuration at x . As x varies, these eigenvectors form an eigenbundle, and the Berry/Wilczek–Zee connection is the natural connection on that bundle.

A clean rank-1 ($U(1)$) reduction requires that the dynamics remains in a single isolated band. If $\Delta\omega_{\text{gap}}(x)$ is the local separation to the nearest competing band and $\Omega_{\text{mix}}(x)$ is the rate of non-adiabatic mode conversion induced by background gradients, then the adiabatic condition

$$\Omega_{\text{mix}}(x) \ll \Delta\omega_{\text{gap}}(x) \quad (28)$$

should hold on the envelope scale. When this fails, a measured “Berry signal” typically reflects mode mixing or beating, and the correct reduction is to retain the relevant multi-dimensional subspace (the Wilczek–Zee description). In the brane model, the pre-stress parameter α and the wavelength regime control branch separation and therefore how well an Abelian gauge description can apply.

5.4 Diagnostic protocol and artifact controls (numerical extraction)

The Berry/Wilczek–Zee connection is meaningful only when it is tied to a specific eigenbundle (or transported subspace) and when the reported observables are gauge-invariant. For numerical experiments on the lattice substrate, the following protocol provides a concrete, reproducible procedure to extract holonomy/curvature and to rule out common artifacts.

Protocol

1. **Carrier isolation / demodulation.** Choose a carrier frequency ω_0 and isolate a narrowband sector (e.g. by band-pass filtering in time or by short-time Fourier methods). Work with a complex envelope representation consistent with Equation (23).
2. **Define the instantaneous polarization object.** At each spacetime point (or each sample along a loop), define either (i) a normalized rank-1 state $|u(x)\rangle$ for an isolated band, or (ii) an n -dimensional subspace projector $P(x)$ (or an orthonormal frame $\mathbf{E}(x)$) for a near-degenerate sector. The definition should be tied to the *local linearized carrier problem* about the slowly varying background at x (cf. Section 5.3).
3. **Measure the adiabaticity/gap indicators.** Estimate a local band gap $\Delta\omega_{\text{gap}}(x)$ (separation to the nearest competing band) and a mode-leakage proxy along the transport path (e.g. growth of energy outside the chosen band/subspace). Use these to empirically validate the adiabatic criterion Equation (28).
4. **Holonomy from overlaps (gauge-invariant).** For a discretized loop $\Gamma : x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_m = x_0$:
 - *Abelian (rank-1):* define links $U_k = \frac{\langle u(x_k) | u(x_{k+1}) \rangle}{|\langle u(x_k) | u(x_{k+1}) \rangle|} \in U(1)$ and report the loop phase $\gamma_\Gamma = \arg(\prod_{k=0}^{m-1} U_k)$.
 - *Non-Abelian (rank- n):* define overlap matrices $M_k = \mathbf{E}(x_k)^\dagger \mathbf{E}(x_{k+1})$ and extract their unitary part (e.g. by polar decomposition) as $U_k \in U(n)$. Report the holonomy matrix $U_\Gamma = \prod_{k=0}^{m-1} U_k$ and gauge-invariant summaries such as $\text{spec}(U_\Gamma)$ and $\text{tr}(U_\Gamma)$.
5. **Curvature by small loops / plaquettes.** Approximate curvature by evaluating holonomy around small loops and dividing by enclosed area (real-space), or use the link-variable Brillouin-zone construction in Section 5.7 (momentum-space).

Artifact controls (required for credibility).

- **Gauge randomization test:** apply arbitrary local phase choices (rank-1) or local $U(n)$ frame rotations (rank- n) and confirm that reported invariants (e.g. γ_Γ , $\text{spec}(U_\Gamma)$) do not change.
- **Resolution/convergence:** verify stability of invariants under refinement of Δt , lattice spacing, and loop sampling density.
- **Adiabatic breakdown demonstration:** intentionally violate Equation (28) and show that the extracted holonomy becomes non-robust (strong dependence on sampling/path or visible subspace leakage), distinguishing geometry from beating.

5.5 Wilczek–Zee connection (non-Abelian)

If the relevant polarization sector is an n -dimensional near-degenerate subspace, choose an orthonormal frame $\mathbf{E}(x) \in \mathbb{C}^{N \times n}$ with columns $|u_a(x)\rangle$ spanning the subspace, so that $\mathbf{E}^\dagger \mathbf{E} = I_n$. Define the Wilczek–Zee connection

$$\mathcal{A}_\mu(x) := i \mathbf{E}(x)^\dagger \partial_\mu \mathbf{E}(x) \in \mathfrak{u}(n). \quad (29)$$

A change of local frame $\mathbf{E} \mapsto \mathbf{E} U(x)$ with $U(x) \in U(n)$ yields the non-Abelian gauge law

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U + i U^\dagger \partial_\mu U, \quad (30)$$

and the curvature (field strength) is

$$\mathcal{F}_{\mu\nu} := \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]. \quad (31)$$

5.6 Axis-aligned triplet phases and coupling-induced mixing ($U(1)^3 \rightarrow U(3)$)

On a cubic lattice substrate it is natural to posit a three-component narrowband carrier triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$ aligned with the lattice axes. Introduce a dimensionless mixing parameter $g_{\text{mix}} \geq 0$ controlling inter-axis mixing. In the lattice substrate, g_{mix} is an effective measure determined by the prestretch α and by the microscopic coupling stencil (notably off-axis/diagonal couplings).

Decoupled limit ($g_{\text{mix}} = 0$). If the three axis channels are dynamically decoupled, each component supports an independent local phase choice and an independent rank-1 Berry connection $a_\mu^{(x)}, a_\mu^{(y)}, a_\mu^{(z)}$. In this limit, it is meaningful (and gauge-invariant) to speak of “the Berry phase of an axis.”

Mixed regime ($g_{\text{mix}} > 0$). For any nonzero mixing, the physical eigenmodes are *linear combinations* of axis labels. Axis-specific phases become basis-dependent and are not gauge-invariant. The correct adiabatic transport is instead a Wilczek–Zee connection on the transported near-degenerate subspace. For an approximately threefold-degenerate carrier subspace, one obtains a $U(3)$ connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ as in Equation (29). It can be decomposed into a trace $U(1)$ part and a traceless $SU(3)$ part,

$$\mathcal{A}_\mu = \frac{1}{3} \text{tr}(\mathcal{A}_\mu) I_3 + \sum_{a=1}^8 A_\mu^a T^a, \quad (32)$$

where T^a are $SU(3)$ generators. This provides a concrete way to express how “axis-aligned Berry phases” transition into a non-Abelian color-like connection once coupling is switched on.

5.7 Berry curvature on a discretized Brillouin zone (link-variable method)

For a lattice substrate, Berry curvature can be computed directly on the discretized Brillouin zone using link variables and plaquettes, closely analogous to lattice-QCD Berry-curvature constructions [6, 12, 16]. Let $|u(\mathbf{k})\rangle$ be a normalized eigenvector (or $\mathbf{E}(\mathbf{k})$ a frame of a degenerate subspace) on a momentum lattice with spacing $\tilde{a}$. In the Abelian case one defines links

$$U_j(\mathbf{k}) = \frac{\langle u(\mathbf{k}) | u(\mathbf{k} + \tilde{j}) \rangle}{|\langle u(\mathbf{k}) | u(\mathbf{k} + \tilde{j}) \rangle|}, \quad (33)$$

and a plaquette

$$U_{ij}(\mathbf{k}) = U_i(\mathbf{k}) U_j(\mathbf{k} + \tilde{i}) U_i^\dagger(\mathbf{k} + \tilde{j}) U_j^\dagger(\mathbf{k}), \quad (34)$$

whose argument approximates the integrated Berry curvature over the plaquette. In the non-Abelian case one uses overlap matrices between frames and projects to $U(n)$ links, then forms the same plaquette structure to obtain a discretized non-Abelian curvature. This is a natural diagnostic bridge between the lattice-substrate hypothesis and QCD-motivated non-Abelian geometry.

5.8 Spinorial holonomy and the 4π aspect

For $n = 2$ (a two-level polarization subspace), holonomy lives naturally in a double cover of $SO(3)$. A closed loop in physical space that induces a 2π rotation of an internal polarization frame can therefore produce a sign change of the transported state, while a 4π loop returns

it to itself. In the present theory this spinorial 4π aspect is attributed to Berry/Wilczek–Zee holonomy of an internal polarization bundle, and is treated as part of the particle-sector structure developed in Section 7.

6 From mode-frame transport to gauge-covariant envelope dynamics

6.1 Multiscale ansatz and retained polarization subspace

Assume an excitation narrowband around a carrier angular frequency ω_0 . Write the substrate displacement as a slowly varying complex envelope multiplied by a local mode frame:

$$\mathbf{u}(x, t) \approx \text{Re} \left[e^{-i\omega_0 t} \sum_{a=1}^n \Psi^a(x, t) \mathbf{e}_a(x, t) \right], \quad (35)$$

where $\Psi(x, t) \in \mathbb{C}^n$ varies slowly compared to ω_0^{-1} and $\{\mathbf{e}_a(x, t)\}_{a=1..n}$ is an orthonormal frame spanning the retained polarization subspace. The frame is not unique: $\mathbf{e}_a \mapsto \sum_b \mathbf{e}_b U_{ba}(x)$ with $U(x) \in U(n)$ represents the same physical subspace.

Projecting the full substrate dynamics onto the retained subspace produces, generically, a gauge-covariant derivative acting on Ψ :

$$\mathcal{D}_\mu \Psi = (\partial_\mu + i \mathcal{A}_\mu) \Psi, \quad (36)$$

where $\mathcal{A}_\mu$ is the Berry (rank-1) or Wilczek–Zee (non-Abelian) connection defined in Section 5. The gauge transformation $\mathbf{E} \mapsto \mathbf{E}U$ induces $\Psi \mapsto U^\dagger \Psi$ and the standard connection transformation law, leaving $\mathcal{D}_\mu \Psi$ covariant.

6.2 Effective action for the slow sector

Under the multiscale separation (single-band or near-degenerate subspace) and smoothness of the polarization bundle, the coarse-grained slow dynamics admits a gauge-invariant effective action built from Ψ and the curvature $\mathcal{F}_{\mu\nu}$:

$$S_{\text{eff}}[\Psi, \mathcal{A}] = \int d^4x \left(\Psi^\dagger \mathcal{K}(\mathcal{D}) \Psi - \kappa \text{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) + \dots \right), \quad (37)$$

where $\mathcal{K}(\mathcal{D})$ is an operator determined by the linearized carrier band (e.g. a Klein–Gordon-type or Dirac-type envelope operator), κ is an emergent stiffness for curvature fluctuations, and “...” denotes higher-order and symmetry-allowed terms. In the Abelian case ($n = 1$) the curvature term reduces to $F_{\mu\nu} F^{\mu\nu}$ with $F_{\mu\nu} = f_{\mu\nu}$. For comparison with electromagnetic conventions one may introduce a dimensionful potential $A_\mu^{\text{eff}} := \eta a_\mu$ (and analogously in the non-Abelian case), so that $F_{\mu\nu}^{\text{eff}} = \partial_\mu A_\nu^{\text{eff}} - \partial_\nu A_\mu^{\text{eff}} = \eta f_{\mu\nu}$. The conversion factor η fixes units and would have to be determined by matching to experiment; it is not introduced as a fundamental coupling in the microscopic substrate equations.

6.3 Triplet envelope sector and $U(3) \simeq U(1) \times SU(3)$ decomposition

If the retained narrowband sector is a three-dimensional near-degenerate subspace (as suggested by an axis-aligned triplet on a cubic lattice), then $n = 3$ and the emergent connection is $\mathcal{A}_\mu \in \mathfrak{u}(3)$. Decomposing into trace and traceless parts,

$$\mathcal{A}_\mu = \underbrace{\frac{1}{3} \text{tr}(\mathcal{A}_\mu) I_3}_{U(1) \text{ part}} + \underbrace{\sum_{a=1}^8 A_\mu^a T^a}_{SU(3) \text{ part}}, \quad (38)$$

suggests a direct route to a “QCD-like” internal gauge structure in the slow-sector EFT: the $U(1)$ part behaves electromagnetism-like, while the traceless part carries non-Abelian self-couplings through the curvature commutator term in $\mathcal{F}_{\mu\nu}$. Crucially, for nonzero inter-axis mixing, statements about “a Berry phase per axis” are not gauge-invariant; the invariant content is the holonomy/curvature of the transported subspace (Wilczek–Zee).

Geometric phases are universal in wave systems [2–4, 11, 15]. What is specific here is the identification of the physically relevant gauge structure with the connection induced by a distinguished narrowband sector of the substrate.

6.4 Dirac-envelope interpretation and the “split”

In standard quantum field theory, the Dirac field ψ simultaneously encodes (i) the internal particle labels (spin, particle/antiparticle) and (ii) the local phase/gauge coupling through the covariant derivative $D_\mu = \partial_\mu + i(q/\hbar)A_\mu$. In the present theory, this mixture is reinterpreted as follows:

- the *particle sector* corresponds to localized solitonic modes with additional internal structure (topology/geometry and its associated labels);
- the *phase/gauge sector* corresponds to the Berry/Wilczek–Zee connection induced by transport of the ordered narrowband polarization subspace.

An effective Dirac-type envelope equation is then understood as a compact coupling of these two sectors. In particular, a “charge”-like coupling parameter is not introduced as a fundamental constant of a new field; it appears as an effective conversion factor between the normalization conventions of the envelope field and the geometric connection extracted from the substrate.

7 Localized modes and self-adjoint eigenproblems

A core goal of the substrate theory is to explain particle-like behavior (localization, discrete spectra, internal “spin/charge”-like structure) as properties of confined wave states rather than point particles.

7.1 Linearization about symmetric backgrounds and self-adjointness

Let $\mathbf{X}_0(x)$ be a time-independent background embedding representing a localized deformation region. We are especially interested in backgrounds that are *spherically symmetric* in the intrinsic brane coordinates, so that invariants depend only on $r = |x|$.

Linearize the equations of motion (13) about $\mathbf{X}_0$ by writing $\mathbf{X}(x, t) = \mathbf{X}_0(x) + \boldsymbol{\eta}(x, t)$ with small perturbation $\boldsymbol{\eta}$. Because the dynamics derives from an action (11), the linearized operator arises from the second variation of the action and is therefore self-adjoint with respect to the inner product induced by the kinetic energy, schematically

$$\langle \boldsymbol{\eta}, \boldsymbol{\zeta} \rangle := \int_{\Omega} \rho_m \boldsymbol{\eta}(x) \cdot \boldsymbol{\zeta}(x) \, d^3x, \quad (39)$$

together with the boundary/regularity conditions appropriate to the chosen domain (periodic cell, decay at infinity, or a large but finite domain with physical boundary conditions). In this setting, orthogonality and completeness of eigenmodes are not “imposed” as ad-hoc constraints: they follow as standard properties of a self-adjoint eigenproblem.

7.2 Angular structure and spherical harmonics

For a spherically symmetric background, the linearized operator commutes with the action of $SO(3)$ on the intrinsic coordinates. Consequently, eigenmodes can be chosen to transform in irreducible $SO(3)$ representations. Operationally, this implies a separation into a radial problem and an angular problem whose eigenfunctions are spherical harmonics $Y_{\ell m}(\theta, \varphi)$. This statement is a symmetry consequence and does not depend on switching the substrate constitutive law: it only requires rotational invariance of the background and of the linearized operator.

In particular, spherical-harmonic mode families provide the natural spatial “skeleton” for localized excitations. Appropriate superpositions (and/or coupling to embedding-direction components) can yield torus-like or ring-like energy density profiles reminiscent of toroidal electron models, but here without assuming literal multi-winding of a photon trajectory.

7.3 Amplitude scaling and energy normalization

The linear eigenproblem fixes mode shapes and frequencies but not the overall amplitude: if $\boldsymbol{\eta}$ is an eigenmode, so is $c\boldsymbol{\eta}$ for any scalar c . Physical amplitudes are selected only when nonlinearities are included and the total energy is fixed. For an electron-like excitation one may impose a rest-energy normalization

$$\overline{E}[\mathbf{X}] = m_e c^2, \quad (40)$$

where $\overline{E}$ denotes the conserved substrate energy associated with (11) and c is the characteristic speed of the carrier branch used to define the effective spacetime coordinates in Section 5. This ties the nonlinear amplitude scale of a localized soliton to a physical energy target.

7.4 Confinement as nonlinear self-guidance

A confining mechanism is required: in a purely linear wave system, localized packets generally disperse or radiate. A useful conceptual benchmark is the toroidal electron proposal attributed to Williamson & van der Mark, which argues that ordinary gravitational curvature is far too weak to confine a Compton-scale photon-like mode and that confinement must arise from a self-consistent, excitation-specific mechanism. We adopt the *confinement requirement* while not committing to a literal double-loop trajectory (see e.g. the review discussion in [8]).

In the present substrate theory, confinement is attributed primarily to the brane’s *constitutive and geometric nonlinearity*: large-amplitude deformations modify the local stress response and therefore the local linearized propagation characteristics of small fluctuations around the deformed state. Equivalently, the soliton “writes” an effective medium for itself (through $g_{ij}(\mathbf{X})$ and W), enabling self-guidance and trapping without introducing a fundamental point charge or external potential.

7.5 Connection to Berry/Wilczek–Zee holonomy and spin

Localized modes can carry internal polarization structure (multi-component standing waves). The Berry/Wilczek–Zee connections of Section 5 can then be evaluated on loops surrounding the localized region or on adiabatic deformations of a localized mode family. In this way, long-range “charge”-like labels are interpreted as far-field holonomy/phase structure, while the spin- $\frac{1}{2}$ 4π aspect is attributed to spinorial holonomy of a two-level polarization subspace.

8 Discussion

8.1 What is fixed by the continuum theory fragment

At the continuum level (treated here as the long-wavelength effective limit of a cubic lattice substrate), the theory specifies: (i) the substrate dynamics via an isotropic hyperelastic action

(11), (ii) the existence of an ordered narrowband sector that supports a controlled multiscale envelope description, and (iii) the geometric connection structure (Berry/Wilczek–Zee) induced by polarization transport. Given these commitments, gauge degrees of freedom are not introduced as independent fundamental fields; they arise as connection data of the ordered carrier sector.

8.2 Minimal-ingredient reconstruction and the role of effective theories

The guiding stance is that established field theories may be *effective* descriptions of stable, repeatable excitation sectors of a deeper substrate. In that reading, the Standard Model is not “wrong”; it is a remarkably accurate macroscopic language for the degrees of freedom that remain after coarse-graining. The present framework aims to reduce the number of primitive ingredients by shifting explanatory weight from “many fundamental objects” to “one object with richer nonlinear and topological behavior”. Concretely, this means: relativistic kinematics is tied to the dominant wave cone of a linear sector, gauge structure is tied to holonomy of an ordered sector, and particle-like behavior is tied to localized nonlinear modes.

8.3 Lattice substrate: ontology, anisotropy, and continuum recovery

The cubic lattice is treated not merely as a numerical convenience but as a candidate *microphysical* substrate. In this reading, the isotropic continuum action serves as a compact *effective* description valid at wavelengths $\lambda \gg a$ (with a the lattice spacing), while the lattice determines the high- k dispersion and the available polarization/mixing structure.

Because cubic symmetry is weaker than full rotational symmetry, the lattice generically produces direction-dependent wave speeds, dispersion, and branch splitting (“birefringence”) at momenta approaching the Brillouin-zone boundary. Two complementary requirements follow: (i) *observational viability* demands that anisotropies at accessible scales be suppressed, and (ii) the theory should explain *why* effective isotropy emerges robustly for composite measuring devices that are themselves excitations of the substrate. Operationally, the paper commits to the controllable regime where the continuum limit is well-defined and anisotropies are parametrically small; the lattice picture then becomes a concrete hypothesis about the UV completion.

8.4 Axis-aligned triplet phases and the coupling-driven transition to non-Abelian holonomy

The “axis-aligned Berry phase” idea is best understood as a *basis-dependent* description of a transported near-degenerate carrier sector. On a cubic lattice, it is natural to organize a narrowband carrier triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ aligned with the lattice axes. If inter-axis mixing is negligible ($g_{\text{mix}} \rightarrow 0$), each component admits an independent rank-1 Berry connection and one may speak meaningfully of three separate $U(1)$ geometric phases.

For any nonzero mixing ($g_{\text{mix}} > 0$), however, the physical eigenmodes are mixtures of axis labels. In that regime, “the Berry phase of the x axis” is not gauge-invariant; the invariant content is the Wilczek–Zee holonomy of the transported subspace and its curvature. This provides a sharp criterion for when axis-wise phase bookkeeping is legitimate (effectively decoupled channels) and when the description must be upgraded to a non-Abelian $U(3)$ connection with an $SU(3)$ traceless component.

8.5 Contact with lattice gauge theory and lattice-QCD Berry curvature diagnostics

Standard lattice QCD uses a lattice primarily as a *regulator* for a continuum gauge theory; in contrast, this framework treats the lattice as *ontic* and interprets gauge structure as emergent connection data of an ordered carrier sector. Despite this difference in interpretation, the *diagnostic machinery* transfers almost verbatim: Berry connections and curvatures can be

computed on a discretized Brillouin zone using link variables and plaquettes, in close analogy to lattice-QCD Berry-curvature constructions [6, 12, 16]. A concrete gauge-invariant extraction protocol and artifact controls are summarized in Section 5.4.

This analogy clarifies a central structural point about the “color” direction. Non-Abelian Berry curvature requires (approximate) degeneracy of the transported subspace; interactions or mixing that lift degeneracy can reduce the effective geometry to an Abelian sector. In this model, the mixing factor g_{mix} plays an analogous role: it controls both the mixing of the triplet and the gap structure that determines whether a non-Abelian adiabatic transport picture is justified. A practical program suggested by this link is: (i) compute $U(1)$ and $U(3)$ Berry curvatures on the momentum lattice for the relevant branches, (ii) compare holonomies under loops in $\mathbf{k}$ -space to “charge” labels of long-range interactions, and (iii) test whether the traceless part exhibits genuinely non-Abelian features (e.g. noncommuting holonomies) in the parameter regime where localized modes and an ordered narrowband sector coexist.

8.6 Gravity as contraction from transverse excitation

The extra embedding dimension is not introduced only to host additional polarizations; it also enforces a specific geometric backreaction: transverse variation changes intrinsic distances (Section 3.8). In the discrete picture this is the Pythagorean spring-length effect, and in the continuum it is already visible in $g_{ij} = h_\star^2 \delta_{ij} + \partial_i u \partial_j u$. Under homogeneous pre-tension, this backreaction produces an inward lateral pull (contraction) around localized transverse excitation. In the intended weak-field regime, the “gravitational field” is therefore identified with a slowly varying, coarse-grained strain/dilation pattern sourced by excitation energy (especially transverse gradient energy), and “gravitational mass” is tied to that stored elastic energy.

What remains open is the quantitative reduction: one would like to derive a Newtonian limit in which an effective scalar potential built from coarse-grained strain or energy density obeys a Poisson-type equation with a universal coupling constant, and then relate that constant to G . On the simulation side, the central check is conceptually simple: create a localized transverse excitation, measure the induced steady contraction field in the surrounding brane, and quantify how it focuses or accelerates long wavelength test packets (or slowly moving localized modes) toward the source.

8.7 de Broglie–Einstein “harmony of phases”

The classical de Broglie picture emphasizes a relation between an internal “clock” (Compton-scale oscillation) and an externally observed matter-wave phase. In the present framework this relation is realized naturally: a localized soliton can support an internal carrier-scale oscillation propagating at the microscopic wave speed c in the effective substrate/branch metric, while the soliton center-of-mass follows a timelike trajectory in the emergent kinematics. The slow envelope phase then encodes the externally observed matter-wave behavior, while the internal carrier provides the phase reference that enables Berry/Wilczek–Zee holonomy.

8.8 Quantum phenomenology without fundamental probability

By construction, the underlying dynamics is deterministic. The probabilistic character of quantum measurements is attributed to the interaction of localized solitons with wave packets that are constrained by Fourier localization and, in the lattice picture, by the finite Brillouin zone (band-limited narrowband carriers cannot be localized arbitrarily sharply), together with threshold-like nonlinear exchange events in which small differences in phase and amplitude can decide whether energy is transferred. In this view, apparent point-like interactions reflect the structure of wave packets and nonlinear exchange, not the existence of fundamental point particles.

8.9 Limitations and open problems

Several gaps remain before the theory fragment could be regarded as a closed replacement for established field theories:

- identifying constitutive laws and parameter regimes that simultaneously support (a) a stable ordered narrowband sector and (b) robust localized solitons,
- deriving the precise effective dynamics of curvature fluctuations of the Berry/Wilczek–Zee connection
- quantifying the lattice-induced anisotropy/birefringence and establishing bounds (or suppression mechanisms) that ensure effective Lorentz/isotropy constraints in accessible regimes,
- formulating the dynamics and physical meaning of the inter-axis mixing factor g_{mix} (including when and how it produces a protected near-degeneracy required for a non-Abelian Wilczek–Zee description), (including whether a Maxwell/Yang–Mills-type term dominates in the relevant regimes),
- clarifying the quantitative mapping between far-field holonomy labels and electromagnetic phenomenology,
- understanding how multi-particle states and apparent entanglement-like correlations arise in a deterministic wave medium.

The present goal is to place the theory on a consistent lattice-to-continuum footing so that these problems can be attacked without ambiguity about the underlying substrate model.

9 Conclusion

We presented a consistent continuum formulation of a deterministic substrate theory in which observable physics arises from waves on a single 3D elastic medium embedded in $\mathbb{R}^4$ (with no built-in preferred directions in the chosen baseline constitutive law), with external time as evolution parameter. The continuum model was stated as an isotropic hyperelastic action built from the induced metric of the embedding, including a simple reference-metric mismatch parameter α that encodes homogeneous pre-stress.

A further geometric consequence of embedding is a transverse–lateral backreaction: localized excitation in the fourth component changes intrinsic distances and can drive a slow in-brane contraction field, providing a qualitative “gravity-like” channel in a weak, slowly varying limit.

Assuming an ordered narrowband carrier sector, we constructed the Berry (rank-1) and Wilczek–Zee (non-Abelian) connections induced by polarization transport and interpreted them as the natural origin of gauge degrees of freedom in a coarse-grained description. We then outlined how localized, particle-like states can be posed as self-adjoint eigenproblems around symmetric backgrounds, with angular structure organized by spherical-harmonic mode families and confinement attributed to geometric/constitutive nonlinearity (self-guidance) rather than to fundamental point charges.

Read as a *minimal-ingredient reconstruction* attempt, the resulting picture ties together: (1) a continuum brane substrate, (2) a transverse–lateral contraction channel as a qualitative origin of gravity, (3) an ordered narrowband phase sector, (4) geometric connection/holonomy as emergent gauge structure, and (5) solitonic fermions with spinorial (4π) holonomy attributed to Berry/Wilczek–Zee transport. The Standard Model and general relativity are treated as empirically validated effective frameworks; the present work aims at a deeper explanatory layer rather than an immediate replacement. A discrete cubic lattice realization serves as the proposed

microphysical substrate; its parameterization is stated in Section 3.2, and the geometric-phase diagnostics are developed in Section 5.

10 Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x, t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	External evolution parameter (time).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g_{ij}^0	Isotropic reference metric: $g_{ij}^0 = \ell_0^2 \delta_{ij}$ (stress-free scale ℓ_0).
$h_\star$	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0/h_\star \in (0, 1]$.
$\hat{g}$	Mixed relative metric: $\hat{g} = (g^0)^{-1}g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\hat{g} - I)$.
λ, μ	Lamé parameters of the StVK baseline stored energy.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
P_A^i	First Piola stress: $P_A^i = \partial W / \partial (\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
ϵ	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = i\langle u \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n -dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ : $\gamma_\Gamma = \oint_\Gamma a_\mu dx^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\text{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
g_{mix}	Effective inter-axis mixing parameter controlling mixing of the triplet channels ($g_{\text{mix}} = 0$ decoupled; $g_{\text{mix}} > 0$ mixed).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.

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